

AI Illustrator

Seven-week build plan — a general illustrator engine, delivered as a product



The deliverable is the engine, not two books. In seven weeks we ship a reusable machine that turns **any** children's-book manuscript into a print-ready, professionally illustrated book — driven by a fine-tuned image model and run from a web app anyone can open. Ella and Bilbo are how we prove it works; the engine is what you keep and run on every title after.

THE PRODUCT

A general engine

ANY MANUSCRIPT, ANY CAST

HOW YOU RUN IT

A client web app

VIEW · REVIEW · GENERATE

HOW WE PROVE IT

Two known books

ELLA & BILBO — THE TEST HARNESS

What we are building

A general illustrator engine — a product, not a project. Any manuscript goes in one end and a print-ready picture book comes out the other, with no code changes per book. The consistency is learned into a fine-tuned model, and the whole run is driven from a web app rather than a developer console.

It reads any manuscript

Any story is split into pages, and its illustration notes are treated as binding instructions — nothing about the book is hardcoded.

It learns any cast

Each main character is fine-tuned into the model as its own trained style, so it is painted naturally and identically on every page.

It composes each page

Backgrounds and characters are assembled with a quiet area reserved for the words before anything is drawn.

It sets the words

Captions and bubbles are placed where they can be read, checked for contrast, and kept word for word to the manuscript.

It builds the book

Front matter, story, back matter and cover are assembled at one trim size and exported ready for the printer.

Anyone can run it

A web app shows every page, the PDF and each character's locked look, and lets you upload a new manuscript and generate a book yourself.

The product is the engine — the two books only prove it

General, not two-book-specific.

Nothing is keyed to Ella or Bilbo; every character, scene and layout is derived from the manuscript at runtime.

Proven on the unseen.

A third manuscript nobody has tuned for is run cold in week 6, so generality is measured, not assumed.

Consistency becomes native.

The model learns each character, so identity is how it paints — not a correction bolted on afterwards.

The cast is a reusable asset.

Trained character weights and stored rules carry into every future title, in this world or a new one.

Runs as a product.

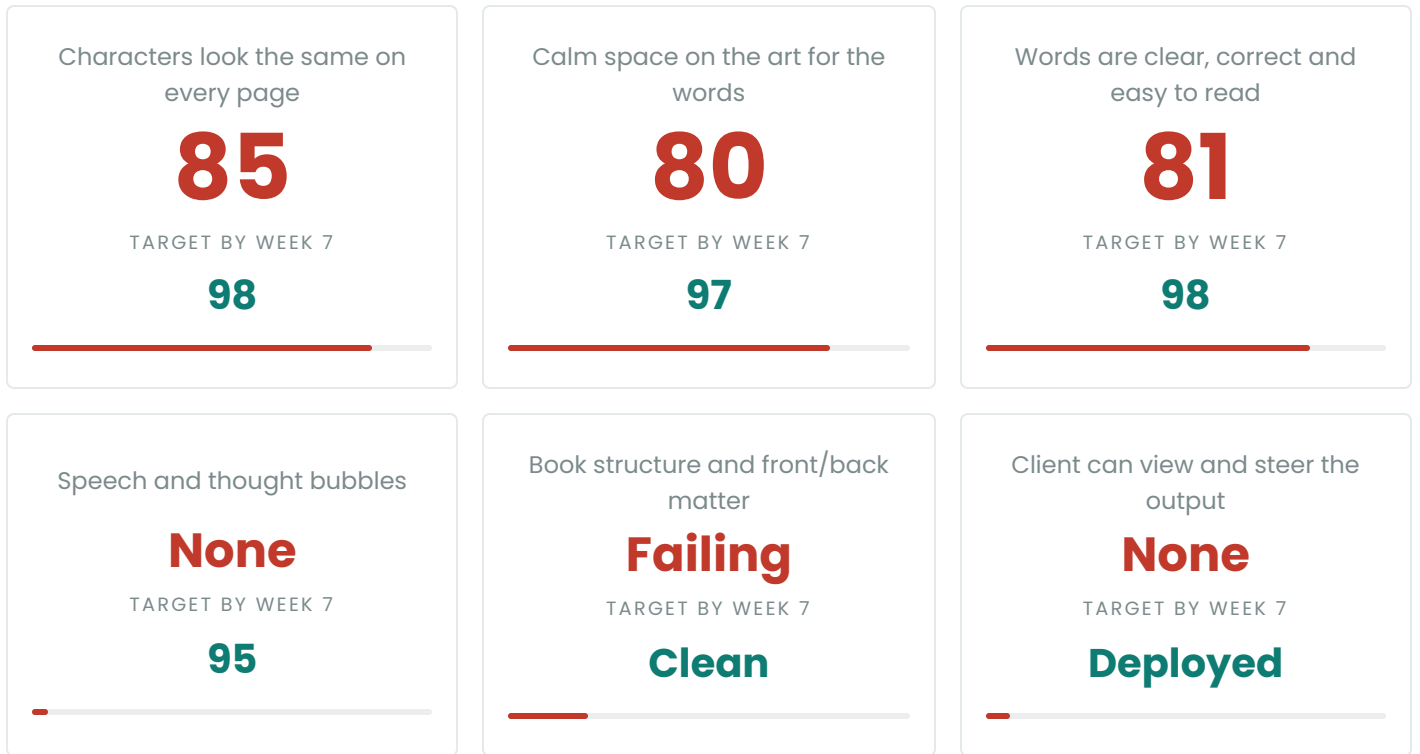
Upload, generate, review, export — a repeatable service, not a one-off render done by hand.

Days, not months.

The next title starts with the engine built, the rules written, and the cast trainable on demand.

Where it stands today

These scores come from a page-by-page review of 39 finished pages, read by eye against the character reference sheets — deliberately not the machine's opinion of itself. They are the starting line every week below is measured against.



What we already have to build on

The full manuscript-to-PDF pipeline exists (parse, copyedit, references, layout, generate, correct, compose, book), with one rule file driving every client complaint into both the generator and the judge. Locked character specs, the categorised client feedback for both books, and the client's three printed interiors as ground truth are all in hand. The layered-compositing engine is kept as a per-page fallback. What is new this cycle is the fine-tuned model and the web app.

The faults we close

Each is a gap in the machine, not a mistake on one page, so fixing it once fixes it for every book that follows. Each is owned by a specific week.

01

Characters change between pages

Size, face and clothing shift page to page. Ella appears twice on one page, Obi loses his hat, Mom's red jersey comes and goes.

FINE-TUNED OUT IN WEEKS 1 TO 3

02

Words land on busy artwork

Text is placed over fences, crowds and shopfronts because no quiet area was planned before the picture was drawn.

ADDRESSED IN WEEK 4

03

Some captions are hard to read

One caption is scrambled into nonsense, and white text over a pale sky has too little contrast to read comfortably.

ADDRESSED IN WEEK 4

04

Working notes leak onto the page

Internal instructions have been printed as if part of the story, and the back cover is still a placeholder.

ADDRESSED IN WEEK 5

05

No speech or thought bubbles

Dialogue is only ever narrated. The bubbles readers expect in this kind of book were never built.

ADDRESSED IN WEEK 5

06

The client cannot see or steer it

The only front end is a broken Streamlit script. There is no way to browse pages, review, or generate a book without a developer.

BUILT ACROSS WEEKS 2 TO 6



The one change that unlocks the rest

Fine-tuning a model per character makes identity and scale the model's native behaviour, so the generator stops fighting drift and the corrector is freed to fix composition instead.

WEEKS 1 TO 3 BUILD ON THIS



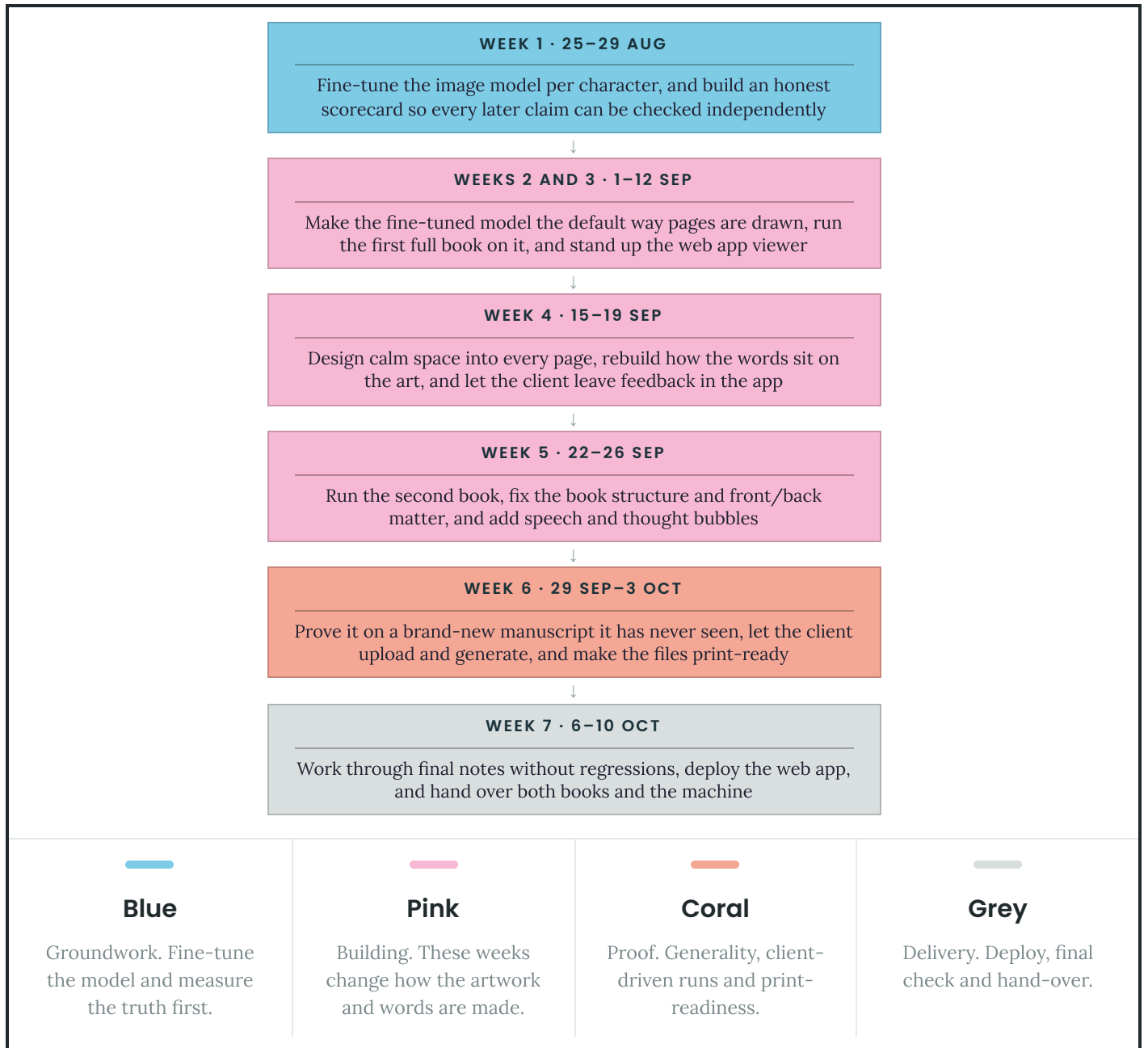
Why the web app runs in parallel

The app is built read-only first, then viewer, feedback, and upload — so every Friday there is something the client can click, and it is shippable at each stage rather than all at the end.

A LITTLE EVERY WEEK

How the seven weeks flow

Every stage upgrades the engine, and nothing later depends on a promise — only on work that has already passed its check.



Week by week

Each week has one headline goal, one measurable outcome, and two client meetings — a Monday review and a Friday demo — and does not close until the outcome is met or the miss is written down.

WEEK 1 · 25–29 AUG · FINE-TUNE

Fine-tune the image model, and measure honestly

- Set up fal.ai access and put the key in place; smoke-test one training call and one generation call before spending on real training.
- Audit the training set: count images and pose variety per character, and flag anyone too thin to train well.
- Build the dataset from each locked reference sheet — 15 to 25 varied poses, angles and expressions per character, each with a unique trigger word.
- Train the first two characters (Ella, Bilbo) on fal.ai, store the trained weights, and wire generation to use them.
- Build the independent scorecard, freeze today's books as the baseline, and A/B three pages: old method vs compositing vs fine-tuned.

OUTCOME

Fine-tuning works for two characters; on the three test pages identity scores 95 or better and is judged more naturally painted than the compositing engine; the scorecard reproduces the hand review within five points and is saved to file.

MON · REVIEW

Present this plan and the baseline scorecard; agree to fine-tune first, with the web app built in parallel.

FRI · DEMO

The same character in several new poses, old method vs fine-tuned, side by side — identity now locked and painted, not pasted.

Fine-tuned model becomes the default; web app stands up

- Train the remaining recurring characters across both books.
- Make the fine-tuned path the default way pages are generated, injecting each character's trigger word and exact locked spec; keep compositing as a per-page fallback.
- Now that identity is native, retune the correction step toward composition — scale, grounding, cropping, location and props — instead of chasing identity.
- Stand up the web app's read-only back end: list books, fetch a book's pages and PDF, and fetch each character's locked references.
- Scaffold the client's provided web design against one real book, and retire the Streamlit path from the client's view.

OUTCOME

A full page renders end to end on the fine-tuned engine, and the web app loads one real book — pages, PDF and character references — at a shareable link.

MON · REVIEW

Sign off the week-1 A/B results; walk through the client's front-end design exactly as we will build it.

FRI · DEMO

A live web link where the client browses a real book — the first time they see the actual thing in a proper app.

First full book on the fine-tuned engine; book viewer finished

- Run the stronger book (Bilbo) end to end on the fine-tuned engine, and regenerate any page that fails the scorecard.
- Extend the scorecard to check clothing persistence and character-size variance across the whole book.
- Handle look-alikes who share a page (Bilbo and Obi) so the model keeps them distinct rather than averaging them.
- Finish the web book viewer — page navigation, spread awareness and PDF download — to the provided design.

OUTCOME

Bilbo is fully on the fine-tuned engine; consistency reaches 95 or better with size varying no more than five percent and no lost hats or drifting clothing; the whole book is browsable in the app.

MON · REVIEW

The fine-tuned engine running on a full book, with the scorecard trend since baseline.

FRI · DEMO

The complete Bilbo book, every page, inside the web app, with the consistency numbers alongside.

WEEK 4 · 15–19 SEP · BUILDING**Calm space and readable words; feedback in the app**

- Decide where the words will go before the picture is drawn, and pass that as a hard constraint asking for open sky, soft ground or plain wall there.
- Fix the fault that scrambled a caption, lock one type size book-wide, and guarantee contrast with adaptive ink and a soft glow, no boxed label.
- Carry the manuscript wording through unchanged, so printed text matches the author's text exactly.
- Add clutter and contrast checks to the scorecard, and redraw any page whose text area is too busy.
- Add per-page comments in the web app that write back into a feedback ledger mapped to our rule codes.

OUTCOME

Calm space and readability both reach 95 or better; no caption is scrambled, cut off or low-contrast; and the client can leave feedback on any page in the app, stored and categorised automatically.

MON · REVIEW

Before-and-after on the worst old pages — words now sitting in real calm space and reading clearly.

FRI · DEMO

The in-app feedback round-trip: a comment on a page becomes a categorised item in the fix queue.

WEEK 5 · 22–26 SEP · BUILDING**Second book, book structure, and bubbles**

- Run Ella end to end on the fine-tuned engine and regenerate any failures.
- Author the real front and back matter — title, dedication, about the author, a fostering and volunteering note, back cover — and add a hard stop that fails the build if any internal note reaches a page.
- Build speech bubbles with a tail to the speaker and thought bubbles as clouds with trailing dots, sized to the text and clear of faces and the calm area.
- Extend the scorecard to check each bubble is readable and points at the right character.

OUTCOME

Ella meets its targets on the fine-tuned engine; book structure comes back clean with no leaked notes and correct matter; and bubbles reach 95 or better wherever dialogue exists, switchable per book.

MON · REVIEW

Ella's regeneration in progress and the newly authored front and back matter.

FRI · DEMO

Both books complete in the web app, with correct structure and the new speech and thought bubbles.

WEEK 6 · 29 SEP–3 OCT · PROOF

Prove generality, let the client generate, go print-ready

- Run a brand-new, unseen manuscript cold — including auto-training its characters — with no code or data edits specific to it, and fix any gap as a general rule.
- Prepare proper print files: correct trim, bleed and safe margins, at print resolution, in print colour, exported for the printer.
- Add the manuscript upload-and-generate flow to the web app: start a run, watch live status, then see the finished book.

OUTCOME

The held-out book scores within five points of the two tuned books on every measure; a manuscript uploaded in the app produces a book; and the print file validates on trim, bleed, resolution and colour.

MON · REVIEW

The held-out result — a book the engine had never seen — and the print-ready export.

FRI · DEMO

The client uploads a manuscript in the app and watches a book generate, hands-off.

WEEK 7 · 6–10 OCT · DELIVERY

Remediation, deploy, and hand-over

- Clear the agreed must-fix list from the in-app feedback, rerunning the full scorecard after each batch so no measure slips.
- Write any new preference into the standing rules, so it is applied automatically from then on.
- Deploy the web app and its back end to a stable, client-accessible URL, and retire Streamlit entirely.
- Run the final scorecard on the print files, and hand over the print PDFs, the source, the trained character weights and a one-page runbook.

OUTCOME

Both books meet their targets and pass the print check; the held-out book stays within five points; the web app is deployed and usable by the client; and you sign off.

MON · REVIEW

The remediation burn-down and a regression scorecard showing nothing slipped back.

FRI · FINAL

The deployed web app, both print-ready books, the held-out proof and the runbook — sign-off.

The progress you should expect to see

Scores are out of 100. Each measure climbs in the week whose task owns it, reaches its target there, and then holds. Structure and the web app are pass-or-fail rather than scored.

Red is still short of target. **Teal** is the week a target is reached and held.

Week	Milestone	Characters	Calm space	Readability	Bubbles	Structure	Web app
0	Starting point, measured by hand	85	80	81	none	failing	none
1	Model fine-tuned, scorecard built	93*	80	81	none	failing	none
2	Fine-tuned engine default, app scaffold	95	82	82	none	failing	read-only
3	First full book on fine-tuned engine	96	84	84	none	failing	viewer
4	Calm space and words rebuilt	96	95	95	none	partial	feedback
5	Second book, structure, bubbles	97	95	96	95	clean	both books
6	Held-out proof, upload, print-ready	98	97	97	95	clean	upload
7	Remediation, deployed, handed over	98	97	98	95	clean	deployed

A note on the numbers

The targets are set to be realistic rather than flattering. Week 1's character score is measured on the three A/B test pages, not a whole book, which is why it carries a star. Past the target the remaining calls are made by eye, which is why the final pass before delivery is a person reading the book rather than a score.

Where the seven weeks go

Fine-tuning takes the largest share, because character consistency is the keystone every later week reuses. The web app runs alongside so every Friday has something to click.



What could slow us down

Each of these is known in advance and has a planned response.

The fine-tune could overfit a thin dataset

Too few or too similar training images and a character only reproduces one pose.

Response. Audit the set in week 1, build 15 to 25 varied images per character, and keep the compositing engine as a per-page fallback.

Training cost or time could run high

Paying to train every character on fal.ai adds up if left unchecked.

Response. Train only recurring characters, batch runs, and track cost per character against the consistency it buys.

Two similar characters could collapse

Look-alikes can blur into each other under one model.

Response. A unique trigger word per character, a paired reference for co-present look-alikes, and a scorecard check that they stay distinct.

The web design could arrive late or change

Building against a design still in flux risks rework.

Response. The back end is design-agnostic and built read-only first against real data, so the design is skinned on top whenever it lands.

A full-book run could surface new faults

A win on three pages may not hold across a whole book.

Response. Always A/B three pages first, and let the scorecard gate every full run before it reaches you.

Late fixes could undo earlier wins

This has happened before, where a fix reopened a problem elsewhere.

Response. The full scorecard is rerun after every batch of fixes, and no measure may slip below its week-5 level.

What you own at the end

The engine and the app come first on purpose, because they are the product — the parts that keep paying on every title after these two. The books sit lower on the list because they are the proof, not the point.

- ✓ **The general illustrator engine**, able to take any new manuscript and return a finished book with no per-book code changes and only light touch-up by hand.
- ✓ **The client web app, deployed**, to upload a manuscript, generate a book, browse pages, preview the PDF, inspect each character's look and leave feedback.
- ✓ **The fine-tune capability**, so any new cast — in these worlds or a brand-new one — can be trained into the model on demand, with the trained weights kept as reusable assets.
- ✓ **The independent scorecard**, so any future book can be measured rather than argued about, proven already on an unseen manuscript.
- ✓ **A short runbook** covering how to run the engine on the next title.
- ✓ The two test books at their week-7 targets — no internal notes on any page, correct front and back matter, one consistent trim size — as the acceptance evidence.
- ✓ Every caption readable, contrast-checked, clear of the centre fold and word for word to the manuscript.
- ✓ Print files at print resolution and colour, with correct bleed and trim, signed off by you.

x

The two meetings each week

Two fixed touchpoints a week keep direction tight and surprises small. The specific thing shown at each meeting is listed under its week in section five.

EVERY MONDAY

Review & kick

We review the weekend's progress and the scorecard, confirm the week's headline goal, and fold in anything from the previous Friday. Roughly thirty minutes.

EVERY FRIDAY

Live demo

We show the week's outcome working — in the web app wherever possible, not as screenshots — and capture your notes straight into the feedback ledger. Roughly thirty to forty-five minutes.

Where your input matters most

Three Fridays carry the most weight: week 3, when the first full book is in the app; week 5, when both books are complete with structure and bubbles; and week 6, when you upload a manuscript and watch it generate. If a Monday or Friday must move, we hold the demo rather than skip it, so no week ever closes unseen.

Read this as the build log for a general AI illustrator engine — a product that turns any manuscript into a print-ready book, run from a web app. The two books are the known-answer test harness that proves the engine, not the deliverable; generality is confirmed cold on an unseen third manuscript in week 6. Starting-point figures come from a page-by-page visual review of what the engine has produced so far, not from its own self-report. Every weekly finish line is measured by the independent scorecard built in week 1, which is deliberately able to disagree with the part of the engine that makes the artwork. Consistency is delivered by fine-tuning a model per character on fal.ai, with the layered-compositing engine kept as a per-page fallback.